

Zonglin Di

CONTACT INFORMATION	4067 MIRAMAR ST # 4831 LA JOLLA, CA 92037-1329 United States	(+1) 858-568-1160 zodi@ucsd.edu
EDUCATION	University of California, San Diego , La Jolla, CA Master of Science, Electrical and Computer Engineering , Dec 2020 to Present • GPA: 3.60/4.0 • Related Courses: Advanced Computer Vision, Visual Learning, Neural Networks & Pattern Recognition, GPU Programming, Statistical Learning I, Reinforcement Robotics Learning Tongji University , Shanghai, China Bachelor of Engineering, Control Science and Engineering , Sept 2014 to July 2019 • GPA: 4.27/5.0 (87.7/100) • Related Courses: Higher Mathematics, Linear Algebra, Possibility Theory & Statistics, Combinatorial Mathematics, Pattern Recognition, Object-Oriented Programming, Database Systems, Operating System, Compiler Principle, Digital Signal Processing, Digital Image Processing, Analog/Digital Circuits, Signal & Systems	
RESEARCH EXPERIENCE	Research Assistant at UBC Dec 2021 to Present • Conduct the medical image registration work using self-supervised methods and reinforcement learning. • Study on algorithms about un/weakly-supervised learning in federated learning • Advisor: Prof. Xiaoxiao Li, Assistant Professor at UBC Research Assistant at UCSD Sept. 2020 to Present • Improve the performance of pose estimation using test-time training and self-supervised learning. The proposed methods can improve the performance by reconstructing another image of the same subject at test-time without annotations. • Advisor: Prof. Xiaolong Wang, Assistant Professor at UCSD Research Assistant of Deep Learning Laboratory June 2018 to Sept 2020 • Perform large-scale roadmap extraction training on 40+ GPUs distributed training and rough labeling to study the limitation of weakly-supervised learning in semantic segmentation • Improve the performance of road extraction using deep learning and GPS data, allowing 5% increase in accuracy and 40% boost in generalization ability for new area prediction • Advisor: Prof. Yin Wang, PhD of University of Michigan Research Assistant of State Lab of Embedded System & Service Computing, Ministry of Education Dec 2014 to Sept 2019 • Developed a novel method to facilitate the detection of Internet addiction disorder (IAD) so as to reduce the subjectivity compared to the traditional methods • Completed a systematic test, showing the new model's capability to identify optimal feature combination in four questionnaires, with the accuracy of IAD labeling reaching 96.2% • Advisor: Prof. Asoke K. Nandi, <i>IEEE Fellow</i> Research Assistant of State Key Lab of Management & Control for Complex Systems, Ministry of Education Sept 2016 to Sept 2019 • Developed a hybrid method for binary and multi-class imbalance classification based on blending support vector machine (SVM) and clustering algorithms and using density peak as the metric	

- Incorporated data distribution and data imbalance simultaneously for improving the imbalance classification of SVM
- Supervisors: Prof. Mengchu Zhou, *IEEE Fellow* & Prof. Qi Kang, *IEEE Senior Member*

PRE-PRINTS
PUBLICATION

1. Shiqiao Meng*, **Zonglin Di***, Siwei Yang and Yin Wang, “Large-scale Weakly Supervised Learning for Road Extraction from Satellite Imagery”
In submission, 2021

CONFERENCE
PUBLICATIONS

1. Miao Hao*, Yizhuo Li*, **Zonglin Di***, Nitesh B. Gundavarapu and Xiaolong Wang, “Test-Time Personalization with a Transformer for Human Pose Estimation”
Advances in Neural Information Processing Systems (NeurIPS), 2021, acceptance rate: 26%
2. Tao Sun, **Zonglin Di**, Pengyu Che, Chun Liu and Yin Wang, “Leveraging Crowdsourced GPS Data for Road Extraction from Aerial Imagery”
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7509-7518, 2019, acceptance rate: 25.2%
3. **Zonglin Di**, Qi Kang, Daogang Peng and Mengchu Zhou, “Density Peak based Pre-clustering Support Vector Machine for Multi-class Imbalanced Classification”
IEEE Conference on System Man and Cybernetics (SMC), pp. 27-32, 2019, acceptance rate: 40%
4. Tao Sun, **Zonglin Di** and Yin Wang, “Combining Satellite Imagery and GPS Data for Road Extraction”
2nd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI), pp. 29-32, 2018, acceptance rate: 40%
5. **Zonglin Di**, Siya Yao, Qi Kang and Mengchu Zhou, “AP-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification”
IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 681-686, 2018, acceptance rate: 40%
6. **Zonglin Di** and Qi Kang, “ADPclust-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification” (Abstract Accepted)
IEEE International Conference on Networking, Sensing and Control (ICNSC), 2018
7. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Detection of Internet Addiction Disorder Based on Personality Questionnaires of Chinese College Students and SVMs”
10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (ICSP-BMEI), pp. 1-6, 2017
8. Xiaoliang Gong, Bozhong Long, Kun Fang, **Zonglin Di**, Yichu Hou and Lei Cao, “A Prediction Based on Clustering and Personality Questionnaire Data for IGD Risk: A Preliminary Work”
12th International Conference on Neural Computation, Fuzzy System and Knowledge Discovery (ICNC-FSKD), pp. 1699-1703, 2016

JOURNAL
PUBLICATIONS

1. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Internet Addiction Disorder Detection of Chinese College Students Using Several Personality Questionnaire Data and Support Vector Machine”
Addictive Behavior Reports, Elsevier, 2019, IF=3.299

TEACHING ASSISTANT	Python Programming (10064701)	Sept 2018 to Nov 2018
	Deep Learning (10068501) Department of Computer Science and Technology, Tongji University Supervised by Prof. Yin Wang	Mar 2019 to June 2019
	Open Source Hardware and Programming (55010501)	Sept 2018 to Jan 2019
	School of Design and Innovation, Tongji University Supervised by Prof. Xiaohua Sun	
START-UP EXPERIENCES	Director of Algorithm Department, Yoyo Technology, Zhejiang June 2019 to June 2020 Our product was about recognizing the fruits and vegetables automatically using deep learning model to improve the checking-out efficiency. Led a team with 3 engineers and designed a complete pipeline from data collection, data labeling, model training, model distillation and model deployment. Improved the performance (classification accuracy) from 70% to 90%. Expanded the category of the classification model to 500+ categories including fruits, vegetables, nuts, meet, etc. Deployed the Efficient-Net v3 ¹ on NVIDIA Jetson Nano after quantization using TensorRT. Received a venture capital from Intel and Hisense.	
INTERN EXPERIENCES	Full-Stack Software Intern in National Instruments, Shanghai April 2018 to April 2019 Established framework of Auto Build System in the Academic Group using React.js, Flask, and JQuery Implemented frontend and backend within one-month timeframe independently Directed by Mr. Dasheng Xu	
	Software Intern in National Instruments, Shanghai	July 2017 to Oct 2017
	Diadem data plug-in from German and Japanese customers using C++ Directed by Ms. Mavis Jin	
	Software Intern in Microsoft, Shanghai	July 2016 to Sept 2016
	Built ASP.NET WebSite and the backend engine Directed by Mr. Tarry Wang and Alex Zhang	
SELECTED AWARDS	Technical Marketing Intern in Jikexueyuan, Beijing	July 2015 to Sept 2015
	Solved the Android issues in the community	
	• The 2nd Prize, 30th National “Challenge Cup” Academic Science and Technology Competition	Nov 2019
	• Special Award, 30th “Challenge Cup” Academic Science and Technology Competition Shanghai	June 2019
	• The Honorable Prize, 2019 Mathematical Contest in Modeling (MCM)	Feb 2019
	• The 2nd Prize, Shanghai Collegiate Maker Competition	Jan 2018
	• The 3rd Prize, China Undergraduate Mathematical Contest in Modeling (Shanghai Division)	Nov 2017
	• Social Practice Scholarship, Tongji University	Sept 2017
	• The 3rd Prize, The 5th “Taidi Cup” Data Mining Contest	Sept 2017
	• The 3rd Prize, The 15th Diligent Design Contest	July 2017
	• The 3rd Prize, The 1st “Huichuangqingchun” Culture & Creativity Project	Mar 2016

¹Tan, Mingxing, and Quoc Le. "Efficientnet: Rethinking model scaling for convolutional neural networks." International Conference on Machine Learning. PMLR, 2019.

SELECTED	• SMC 2019, Bari, Italy	Oct 2019
PRESENTATIONS	• CVPR 2019, Long Beach, USA	June 2019
	• GeoAI 2018, Seattle, USA	Nov 2018
	• SMC 2018, Miyazaki, Japan	Oct 2018
	• ICNSC 2018, Zhuhai, China	Mar 2018
	• ICSP-BMEI 2017, Shanghai, China	Sept 2017
	• ICNC-FSKD 2016, Changsha, China	Aug 2016
	• Excellent Undergraduate, Tongji University	Sept 2015, 2016, 2017
SELECTED SOCIAL	IEEE CVPR 2019 Student Volunteer	June 15 to June 21, 2019
EXPERIENCE	Helped with coordinating session transitions for 5,000 conference attendees.	
	Google Developer Group	Dec 2016 to Sept 2018
	Invited to Google Developer Days in 2016 and 2017 for over 2,000 attendees	
	Organized the Google DevFes 2017 for over 200 attendees	
	Organized the Google TensorFlow Technical Salon on the application of deep learning in computer vision and natural language processing	
	Unity Unite, Shanghai and Beijing	May 2017 and May 2018
	Keynote and Tech Conference Track Support	
REVIEWER	• The 2nd International Conference on Mechanical, Electric and Industrial Engineering	
EXPERIENCE	• ACM SIGSPATIAL 2019 (External)	
	• WACV 2022	